# Thursday Warm-up: Vector Algebra Review

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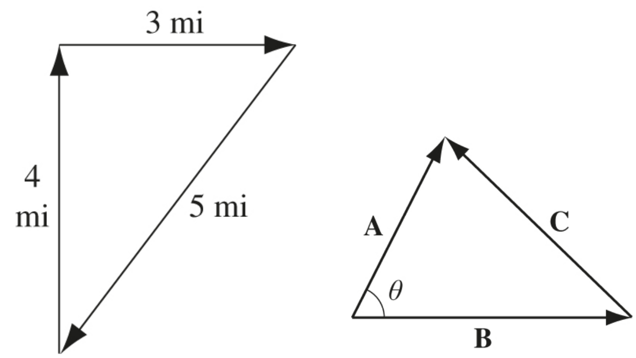


Figure 1: (Left) A three-stage journey, expressed as vectors. (Right) In this triangle, $\mathbf{A} - \mathbf{B} = \mathbf{C}$.

## 1 Review: Vector Algebra

1. (a) What is the total distance traveled in Fig. 1 (left), assuming the traveler begins at the origin, and goes North by 4 miles? (b) At the end of the second stage, what is the location of the traveler? Give both the coordinates and the angle with respect to the x-axis.

2. Consider Fig. 1 (right). Let $\mathbf{A} - \mathbf{B} = \mathbf{C}$. (a) Prove the *law of cosines* by computing $\mathbf{C} \cdot \mathbf{C}$. (b) Suppose $\theta = 90$ degrees. What is the area of the triangle?

3. (a) Show geometrically that the cross-product $\mathbf{A} \times \mathbf{B}$ in Fig. 2 is equal to the area of the parallelogram. (b) In what direction is the cross-product?

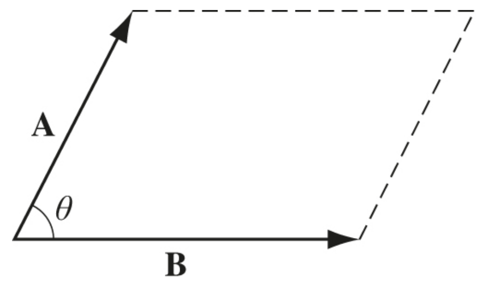


Figure 2: The magnitude of the cross-product of $\mathbf{A}$ and $\mathbf{B}$ is equal to the area of the parallelogram.

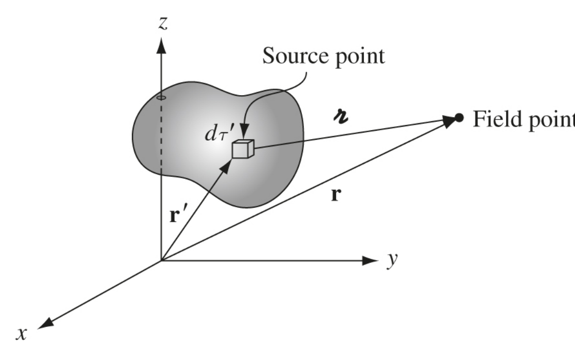


Figure 3: The gray region represents a quantity of electric charge, located within a 3D Cartesian coordinate system.

## 2 Displacement Between Charge and Field Point

1. Consider Fig. 3. The vector position of a piece of electric charge relative to the origin of the Cartesian coordinate system is $\mathbf{r}'$, and the vector position of the point where we observe the field produced by the charge is $\mathbf{r}$. (a) Show that the displacement between the charge point and field point is $\mathscr{r} = \mathbf{r} - \mathbf{r}'$. (b) Calculate the magnitude of $\mathscr{r}$ by taking the dot product $\mathscr{r} \cdot \mathscr{r}$, and then taking the square root. (c) Suppose $\mathbf{r} = 5\hat{y} + 4\hat{z}$ cm, and $\mathbf{r}' = 2\hat{y} + 2\hat{z}$ cm. What is the distance from the charge point to the field point?